Making character consistency measurable

Iteration 01: portrait retention tested

Scoring implemented.
Mixed result; validate next.

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What existed and what we added

Existing before this work	Added during this work
Story, scene prompts, canonical portraits and reference-conditioned editing	SCRFD and ArcFace scoring of all nine saved passes, with crops and candidate matches
Saved passes, prompts and a technical camera-roll viewer	Story-first website with face and object score comparisons
Proposed identity-retention experiment	Controlled portrait-reference comparison with preserved images, prompts and scoring evidence

Existing Maya pilot: five scenes, eight references, nine scene passes.

Sources: proposal pp. 1–3; evaluation documentation; rating guide.

How we measure face consistency

1. SCRFD finds faces. It locates faces and landmarks such as the eyes and nose. We use these landmarks to crop and align each face consistently.

2. ArcFace describes identity. A pretrained recognition network converts each face into a compact set of numbers. Its training encourages images of the same person to have similar representations.

3. Compare with the portrait. We compare those numbers with the character's reference using cosine similarity. The score ranges from -1 to 1. Higher means a closer model match.

Why these tools? Efficient face detection and identity-focused comparison, with pretrained models that run locally and score every pass consistently.

Human review checks whether score changes reflect identity changes.

Sources: SCRFD paper; ArcFace paper; local evaluator.

The two models at a glance

SCRFD Face detection

Finds faces in an image and estimates facial landmarks, including the eyes and nose.

In our pipeline: locate, crop and align each face before scoring.

ArcFace Face recognition

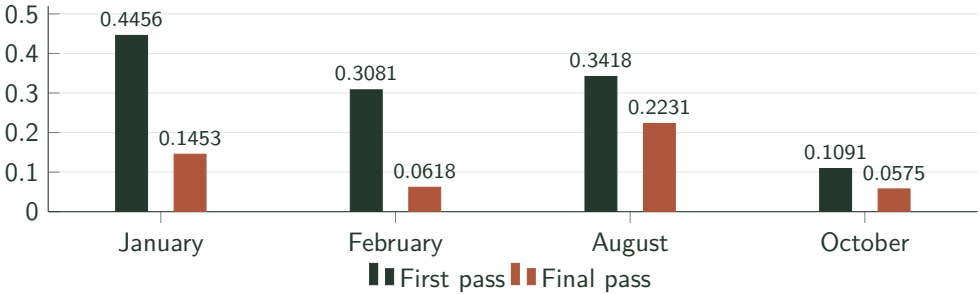
Encodes an aligned face as a numerical representation designed to distinguish identities.

In our pipeline: compare the candidate and reference representations using cosine similarity.

Sources: SCRFD paper; ArcFace paper.

Historical pilot: Maya's score fell in four edited scenes

First pass versus final pass. Higher similarity means a closer model match.

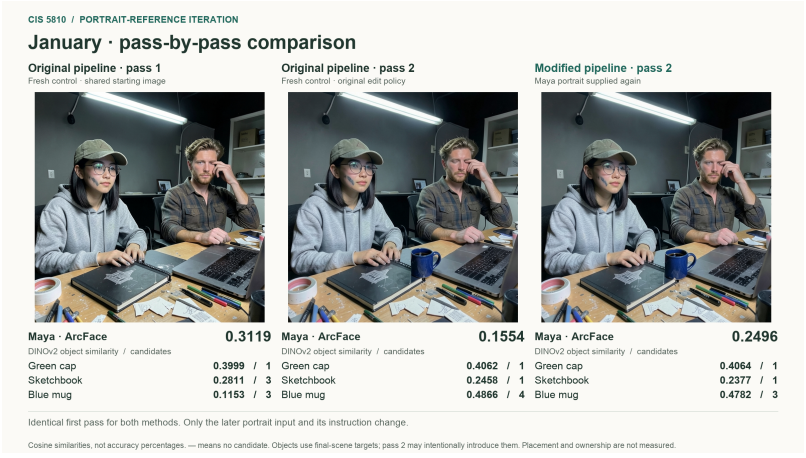


May: 0.3098, single pass. No before/after comparison.

This historical signal motivated the controlled portrait-retention test.

Source: saved ArcFace results. Scores compare assigned candidates with Maya's canonical portrait.

January: a controlled comparison of the portrait edit



Same first-pass image in both conditions; only the portrait input and its instruction changed. Source: Iteration 01 evidence package.

Portrait retention partly recovered face similarity

Scene	Shared pass 1	Original pass 2	Portrait pass 2
January	.3119	.1554	.2496
February	.1973	.0910	.1743
August	.1002	.0726	.1106
October	.1277	.1284	.2039*

Same-candidate scenes: mean .1063 → .1782 after the change; still below the shared first-pass mean (.2031). October is a matcher failure: it selected another person. The intended person's score fell .1284 → .0696.

ArcFace cosine similarities, not accuracy percentages. Three-scene subset is post-hoc. One five-scene run; no repeated trials or confidence intervals. Source: analysis and limitations.

Object resemblance stayed essentially unchanged

Matched edited-scene objects

Same 10 appearances with candidates in both conditions; shared May excluded.

.3064 → .3045
Mean DINOv2 cosine similarity

Change: −0.0020.

What the score cannot tell us

It measures resemblance to one reference view. It does not verify object placement, owner, count or state.

In February, the mug moved from the requested radiator to Maya's hand; its score fell .5153 → .3317.

Localization: Grounding DINO. Similarity: DINOv2 on selected crops. Source: object results and caveats.

Human judgments help interpret the scores

Automated comparisons are reproducible. The package preserves images, prompts, manifests, model outputs and matched scores.

Human review is still needed. Check identity, object count, placement and state, plus setting and action.

Uncertainty stays visible. Allow unclear judgments. Retain raw votes, disagreements and prior-exposure disclosures.

No independent human ratings or repeated generation trials are included yet.
Keep portrait retention experimental; the evidence shows partial recovery and meaningful failure cases.

Source: study limits; rating guide.

Decision: keep portrait retention experimental

Finding: resupplying Maya's portrait improved assigned face similarity in three matched scenes, with an October wrong-person failure and no average object-resemblance gain.

What to keep	What to test next
Portrait input as a promising experimental option	Direct the edit to the intended person's region; run repeated generations and blinded paired human review

Iteration 01 evidence: full results, decisions and limitations.